A CERTIFIED COMPUTATION OF $\Lambda \le 0.169$ WITH A LONG MOLLIFIER, AND AN INDEPENDENT CHECK OF EARLIER BOUNDS

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Abstract. Using the upper-bound criterion of Polymath15 [1] and the verification of the Riemann hypothesis to height 3,000,175,332,800 by Platt and Trudgian [2] as the only external computation, we certify in ball arithmetic that the de Bruijn–Newman constant satisfies $\Lambda \le t_0 + \frac{1}{2}y_0^2 = 0.169$, with $t_0 = 0.1658$, $y_0 = 0.08$. The new ingredient is a b-weighted mollifier of Selberg–Levinson type, $\sum_{d\le 299,\ d \text{ squarefree}} \mu(d) \prod_{p\mid d} b_p\, P(\log(300/d)/\log 300)\, d^{-s}$, used in the hardest range of the canopy. It is combined with the improved triangle inequality of [1] and with constants taken directly from [1, Thm. 1.3]. Along the way we reproduce Polymath15's $\Lambda \le 0.22$ independently, record several arithmetic slips in that paper (none affecting its result), and reproduce the first 2,000 windows of the recent computer-assisted bound $\Lambda \le 0.1787854$ [4] exactly. All code, coefficients and logs are public [7]. The result has not been peer reviewed; we ask the reader to check the items in §1.

1. Summary and status

Claim	Status
$\Lambda \le 0.169$ ($t_0 = 0.1658$, $y_0 = 0.08$)	All hypotheses of [1, Thm. 1.2] certified in arb ball arithmetic (§3).
$\Lambda \le 0.173,\ 0.176,\ 0.18$	Earlier certified rows of the same pipeline (Euler mollifiers); superseded.
Polymath15's $\Lambda \le 0.22$	Reproduced independently; slips listed in §4.
Gomila's $\Lambda \le 0.1787854$ [4]	First 2,000 finite windows reproduced digit for digit (§5).
Limit of the method at this height	Float scouting: ≈ 0.168; 0.165 is out of reach (§6).

Everything rests on [1, Thms. 1.2, 1.3], the derivative and monotonicity lemmas of [1, §8], and [2]. Nothing here bears on the Riemann hypothesis beyond the equivalence RH $\iff$ $\Lambda = 0$ [3]. Allowed reading: *subject to those inputs, the Polymath criterion yields* $\Lambda \le 0.169$.

2. Setting

With $\Phi(u) = \sum_{n\ge 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}$ put $H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du$, so $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$. We use, as published: the criterion [1, Thm. 1.2] (RH to height $X/2$, a zero-free canopy $y_0 \le y \le 1$ for $x \ge X$ at $t = t_0$, and a barrier on $[X, X+1] \times [y_0, 1] \times [0, t_0]$ give $\Lambda \le t_0 + \frac{1}{2}y_0^2$), and the effective approximation $H_t/B_t = f_t + O_{\le}(e_A + e_B + e_{C,0})$ of [1, Thm. 1.3], with $f_t = \sum_{n\le N} b_n n^{-s_*} + \gamma \sum_{n\le N} n^y b_n n^{-\overline{s_*}-\kappa}$, $b_n = e^{\frac{t}{4}\log^2 n}$.

3. The certified computation of $\Lambda \le 0.169$

Parameters. $t_0 = 0.1658$, $y_0 = 0.08$. Barrier centre $X_0 = 6{,}000{,}347{,}387{,}366$ (so $X = X_0 - \frac{1}{2}$ satisfies $X/2 = 3{,}000{,}173{,}693{,}682.75 \le T_{\mathrm{PT}}$), chosen to make $|\prod_{p\le 11}(1 - p^{-(1-iX_0)/2})^{-1}| \approx 32$

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large. $N_0 = 691008$ is constant on the barrier. Canopy split: region A $N \in [691008, 1243814]$, region B $N \in [1243814, 5 \cdot 10^8]$, tail $N \geq 5 \cdot 10^8$.

Mollifiers. Region A uses the long mollifier $M(s) = \sum_{d \in S} \lambda_d d^{-s}$, $S = \{d \leq 299 : \ \mu(d) \neq 0\}$ (183 terms), $\lambda_d = \mu(d) \prod_{p|d} b_p\, P(x_d)$, $x_d = \log(300/d)/\log 300$, $P(x) = x - 1.805(x^2 - x)$, frozen as exact decimal coefficients (file `stage169/mollifier_y300.txt`). Any real coefficients give a valid mollifier: the argument principle is applied to $M \cdot H_t/B_t$, and the error term is multiplied by the certified $\max |M| \leq 1 + \rho$, $\rho = \sum_{d \geq 2} |\lambda_d| d^{-\sigma}$, with $|M| \geq 1 - \rho$ and $|\arg M| \leq \arcsin \rho$ used on the other edges. Region B uses the Euler factor $\prod_{p \leq 5}(1 - b_p p^{-s})$.

Lower-edge bound. For each block $[N_-, N_+]$ we bound $\operatorname{dist}(M f_t, (-\infty, 0])$ uniformly in T by the improved triangle inequality of [1, §8],

$$F = 1 - c\alpha_1 - \sum_{n \geq 2} \max\big(|\beta_n - c\alpha_n|,\ r|\beta_n + c\alpha_n|\big) n^{-\sigma} - Z,$$

with coefficients $n/d \leq N_-$ kept exactly and those with $N_- < n/d \leq N_+$ counted in absolute value, so that the bound holds for every N in the block. Constants come directly from [1, Thm. 1.3, (20)–(21)]: $\sigma = \frac{1+y_0}{2} + \frac{t_0}{2} \log N_- - 10^{-9}$ and $|\gamma| n^y \leq e^{0.02 y_0}(1 + 10^{-9})(n/N)^y$, in place of the convenience constants 0.001 and 1.03 of [1, §8].

Results (arb ball arithmetic).

Piece	Certified result	Required
(i) RH to $X/2$	[2]	external
(iii) Barrier, $t \in [0, 0.1658]$	24 time slices, no gaps; 1327 steps; $\min \lvert f_t \rvert \geq 1.324$; winding 0	$> 7.03 \times 10^{-4}$ + slack
(ii) Lower edge, A (115 blocks)	$\min F \geq 0.0136527$ (first block)	$> 7.67 \times 10^{-7}$
(ii) Lower edge, B (153 blocks)	all positive	$> 6.54 \times 10^{-7}$
(ii) Upper edge $y = 1$	dist ≥ 0.01383 (A), 0.1834 (B)	$\leq 7.67 \times 10^{-7}$
(ii) Left edge (A)	$\min \lvert f \rvert = 1.4850$, dist ≥ 0.0577	7.67×10^{-7}
(ii) Internal edge x_s = 19,441,111,665,846	$\min \lvert f \rvert = 1.1811$; dist ≥ 0.1326 (A), 0.6307 (B)	$\leq 2.8 \times 10^{-7}$
(ii) Tail $N \geq 5 \cdot 10^8$, right edge	$\lvert f \rvert \geq 0.1117$; dist ≥ 0.0575	$\sim 10^{-12}$

The tail uses [1, Lemma "largen"] twice (A in arb); monotonicity of the B-term in N is proved: its four constituent log-derivatives are decreasing in $\log N$ and are ≤ -0.88 at $\log N_1$. The internal vertical edge lets the argument principle close separately on the two regions with different mollifiers. The large sums run on Cloud Run (881 tasks, 108.6 vCPU-hours) with the C kernel `cloud/dbn.c` (FLINT 3.0.1); block sums split across tasks are recombined in exact rational arithmetic from upward-rounded partial bounds.

4. Reproduction of Polymath15 and errata

At $t_0 = y_0 = 0.2$, $X_0 = 6 \times 10^{10} + 83952$ we re-derived [1, §8] and confirm $\Lambda \leq 0.22$, noting: (1) in the bound $e_A + e_B \leq 3.444 \times 10^{-9}$ the factor written 1.023 should be ≈ 2.023 and the exponents $0.5999, 0.80005$ should be $0.4001, 0.79995$ (corrected: 1.36×10^{-8}; the total $\leq 1.25 \times 10^{-3}$ stands); (2) in claim (c), $A = 1.9063$ rather than ≤ 1.88, so $|f_t - 1| \leq 0.9804$; the claim still holds with margin; (3) the claim (b) coefficients are written without the truncation $n/d \leq N$; done exactly, the interval minima are $0.0202, 0.0408, 0.0867, 0.0525$ (stated: $0.0263, 0.0470, 0.093, 0.060$), all far above 2.15×10^{-3}; (4) the barrier time-step condition appears to contain a typo ($D_t|t' - t|/(2n)$ for $D_t|t' - t|$). We also checked [1, Thm. 1.3] against an independent rigorous evaluation of H_t at 18 points ($x \leq 1000$).

5. Independent reproduction of the 0.1787854 pipeline

At commit `a74738de` of [4] we compiled the finite-lane producer with the published flags and reproduced the first 2,000 windows ($N = 690988, \dots, 692987$): all 2,000 floors agree with the logged values digit for digit (minimum 7.91366×10^{-7} at $N = 690988$). We note that the certified sweep is compiled with the plain triangle weight $|b| + \gamma|a_0|$ (`-DTRIANGLE_WEIGHT`), although the same source also implements the improved weight. In float scouting at that row, Euler-11 gives a hardest-window bound $\approx 1.6 \times 10^{-5}$ with the plain weight and ≈ 0.136 with the improved one; this, together with the long mollifier, accounts for the difference in reach.

6. What limits the method

Float scouting at the hardest window ($N = N_0$), same height (non-rigorous):

Mollifier (improved weight, tight constants)	approximate wall	comment
Euler-5	0.1755	
Euler-11	0.1717	per-prime gains $\approx p^{-\sigma}$
Euler with multi-tap factors per prime	0.1712	only $+0.011$ in F
b-weighted long mollifier, $y \approx 300$	≈ 0.168	gains saturate by $y \approx 300$–1000

At $\Lambda = 0.165$ the best floor found is ≈ -0.07 and at 0.160 it is ≈ -0.21. The Theorem 1.3 error terms ($\sim 5 \times 10^{-7}$) are never the binding constraint; the binding constraint is the uniform lower bound for the mollified Dirichlet polynomial near N_0, which only a larger verification height or a new lower-bound technique can move.

7. Limitations

The result is conditional on [1, 2] as published. Parameter selection used float scouting; every certified inequality is decided on arb ball endpoints. It has not been peer reviewed. Two recent claims, $\Lambda < 0.16$ and $\Lambda < 0.1595$, rest on a different (collision-exclusion) argument whose key numerical leaves are not published; we neither confirm nor refute them (Appendix A).

8. Reproducibility

Repository [7]: `cloud/dbn.c` (certified kernels: blocks with arbitrary mollifier, barrier, edges, left and internal edges), `cloud/tasks.json` (exact work split), `results/cert169/` (all task logs), `stage169/` (mollifier coefficients and edge certificates), and the float scouting scripts.

Appendix A. Notes on the certificate for $\Lambda < 0.1595$

The certificate [5] argues by collision exclusion on $t \in [0.1595, 0.1875]$, anchored at its author's $\Lambda \le 3/16$. Its exterior module asserts $|u_x| > 1/400000$ at all real zeros with $x \ge 6{,}000{,}000{,}185{,}827$, supported by 42 numerical leaves whose outputs are not included and by an analytic bound for $N \ge 2{,}299{,}999$. A spot check of 2,655 real zeros in six windows between 6×10^{12} and 6.6×10^{13} at $t = 0.1595$ found $\min |\frac{d}{dx}(H_t/B_t)| = 7.71$ at the zeros, consistent with the claim; a sample cannot confirm it. Publishing the leaf outputs and the derivation of the infinite-branch bound would allow independent verification.

Acknowledgements

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References

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